

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:-06

AIM : Decision Making under Uncertainty

- Develop a decision tree model to make business decisions considering uncertainties and associated probabilities at each decision point.
- Apply the concept of expected value to evaluate different decision alternatives and select the optimal one.

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T128_P6.py - C:\Users\Rani\Downloads\T128_P6.py (3.13.7)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt

print("Name :- Rani Chavre")
print("Roll no.:- T128")
print("AIM :- Decision Making under Uncertainty")
df = pd.read_csv("Rani_Decision_Making_Dataset.csv")

print("\n===== DATASET =====")
print(df)

df["Weighted_Profit"] = (
    df["Demand_Probability"] * df["Profit"]
)

print("\n===== WEIGHTED PROFIT =====")
print(df)

expected_value = (
    df.groupby("Product")["Weighted_Profit"]
    .sum()
    .reset_index()
)

expected_value.columns = ["Product", "Expected_Value"]

===== RESTART: C:\Users\Rani\Downloads\T128_P6.py =====
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

===== DATASET =====
Product Demand_Level Demand_Probability Profit
0 Laptop High 0.30 450000
1 Laptop Medium 0.45 320000
2 Laptop Low 0.25 180000
3 Smartphone High 0.35 400000
4 Smartphone Medium 0.40 300000
5 Smartphone Low 0.25 160000
6 Tablet High 0.25 280000
7 Tablet Medium 0.50 220000
8 Tablet Low 0.25 120000
9 Headphones High 0.30 180000
10 Headphones Medium 0.45 130000
11 Headphones Low 0.25 80000
12 Smartwatch High 0.35 210000
13 Smartwatch Medium 0.40 150000
14 Smartwatch Low 0.25 90000

===== WEIGHTED PROFIT =====
Product Demand_Level Demand_Probability Profit Weighted_Profit
0 Laptop High 0.30 450000 135000.0
1 Laptop Medium 0.45 320000 144000.0
2 Laptop Low 0.25 180000 45000.0
3 Smartphone High 0.35 400000 140000.0
4 Smartphone Medium 0.40 300000 120000.0
5 Smartphone Low 0.25 160000 40000.0
6 Tablet High 0.25 280000 70000.0
7 Tablet Medium 0.50 220000 110000.0
8 Tablet Low 0.25 120000 30000.0
9 Headphones High 0.30 180000 54000.0
10 Headphones Medium 0.45 130000 58500.0
11 Headphones Low 0.25 80000 20000.0
12 Smartwatch High 0.35 210000 73500.0
13 Smartwatch Medium 0.40 150000 60000.0
14 Smartwatch Low 0.25 90000 22500.0

===== EXPECTED VALUE =====
Product Expected_Value
0 Laptop 144000.0
1 Smartphone 140000.0
2 Tablet 110000.0
3 Headphones 58500.0
4 Smartwatch 73500.0
5 Tablet 70000.0
6 Smartphone 120000.0
7 Headphones 54000.0
8 Smartwatch 60000.0
9 Laptop 135000.0
10 Smartwatch 22500.0
11 Headphones 20000.0
12 Tablet 45000.0
13 Smartphone 40000.0
14 Tablet 30000.0

optimal = expected_value.loc[
    expected_value["Expected_Value"].idxmax()
]

print("\n===== OPTIMAL DECISION =====")
print("Best Product:", optimal["Product"])
print("Expected Value: ₹",
    round(optimal["Expected_Value"], 2))

plt.figure(figsize=(14, 8))

# Starting point
plt.scatter(0, 0, s=1200)

plt.text(
    0, 0,
    "BUSINESS\nDECISION",
    ha="center",
    va="center",
    fontsize=10
)

products = df["Product"].unique()
for i, product in enumerate(products):
```

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